

2014 A Level Paper 1 Suggested Solutions

1	Suggested Solutions
(i)	$f(x) = \frac{1}{1-x}$ $f^2(x) = f[f(x)]$ $= \frac{1}{1 - \left(\frac{1}{1-x}\right)}$ $= \frac{1}{\left(\frac{-x}{1-x}\right)}$ $= \frac{1-x}{-x}$ $= \frac{x-1}{x}$ Let $y = \frac{1}{1-x}$ $y - xy = 1$ $xy = y - 1$ $x = \frac{y-1}{y}$ $f^{-1}(x) = \frac{x-1}{x}$ $\therefore f^2(x) = f^{-1}(x) \text{ (shown)}$
(ii)	$f^3(x) = f[f^2(x)]$ $= ff^{-1}(x) \quad (\text{Since } f^2(x) = f^{-1}(x) \text{ from (i)})$ $= x$ <u>Alternative Method</u> $f^3(x) = f[f^2(x)]$ $= \frac{1}{1 - \left(\frac{x-1}{x}\right)}$ $= \frac{1}{\left(\frac{1}{x}\right)} = x$

2	Suggested Solutions
	$x^2y + xy^2 + 54 = 0 \quad \text{----- (1)}$ Differentiate wrt x, $2xy + x^2 \left(\frac{dy}{dx} \right) + y^2 + x \left(2y \frac{dy}{dx} \right) = 0$ Since gradient $= -1 \Rightarrow \frac{dy}{dx} = -1$, $2xy + x^2(-1) + y^2 + 2xy(-1) = 0$ $-x^2 + y^2 = 0$ $y^2 = x^2$ $y = \pm x$ $\therefore y = x \text{ or } y = -x$ Sub $y = -x$ into (1), $x^2(-x) + x(-x)^2 + 54 = 0$ $-x^3 + x^3 + 54 = 0 \text{ has no solution.}$ Sub $y = x$ into (1), $x^2(x) + x(x)^2 + 54 = 0$ $x^3 + x^3 + 54 = 0$ $2x^3 = -54$ $x^3 = -27$ $x = -3$ when $x = -3, y = -3$ $\therefore$ There is only one such point, with coordinates $(-3, -3)$.

3	Suggested Solutions
(i)	Since $\mathbf{a} \times \mathbf{b} = \mathbf{0}$, $\mathbf{a}$ or $\mathbf{b}$ are zero vectors or $\mathbf{a}$ and $\mathbf{b}$ are parallel to each other. <u>Alternative method</u> $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ $ \mathbf{a} \times \mathbf{b} = 0$ $ \mathbf{a} \mathbf{b} \sin \theta = 0$ $\therefore \mathbf{a} = 0, \mathbf{b} = 0 \text{ or } \sin \theta = 0 \Rightarrow \theta = 0^\circ \text{ or } 180^\circ$ Hence, either $\mathbf{a}$ or $\mathbf{b}$ or both are zero vectors or $\mathbf{a}$ and $\mathbf{b}$ are parallel to each other.
(ii)	Since $\mathbf{n}$ is not a zero vector, $\mathbf{n}$ must be parallel to $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

	$\left \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right = \frac{1}{\sqrt{1^2 + 2^2 + (-2)^2}} = \frac{1}{3}$ $\therefore \mathbf{n} = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \text{ or } -\frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$
(iii)	Let the required acute angle be q. $\cos \theta = \frac{\left \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right }{\left \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right \left \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right }$ $= \frac{ -2 }{3} = \frac{2}{3}$

4	Suggested Solutions
[Out of Syllabus (9758)]	

Qn 5	Suggested Solutions
(i)	$z^2 = (1+2i)^2$ $= 1 + 4i + (2i)^2 = -3 + 4i$ $\frac{1}{z^3} = \frac{1}{(1+2i)(-3+4i)}$ $= \frac{1}{-11-2i}$ $= \frac{-11+2i}{11^2+2^2}$ $= \frac{-11+2i}{125} = -\frac{11}{125} + \frac{2}{125}i$ • $\frac{1}{-11-2i} = \frac{1}{-11-2i} \times \frac{-11+2i}{-11+2i}$ • Recall that for $z = x+iy$, $zz^* = x^2 + y^2$.
(ii)	$pz^2 + \frac{q}{z^3} = -3p + 4pi - \frac{11}{125}q + \frac{2}{125}qi$ $= -3p - \frac{11}{125}q + \left(4p + \frac{2}{125}q\right)i$ For $pz^2 + \frac{q}{z^3}$ to be real, $4p + \frac{2}{125}q = 0$. Since $pz^2 + \frac{q}{z^3}$ is purely real. $\frac{2}{125}q = -4p$ $\therefore q = -250p$

	$pz^2 + \frac{q}{z^3} = -3p - \frac{11}{125}(-250p)$ $= 19p$
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Qn 6	Suggested Solutions
(a)(i)	Out of Syllabus (9758)]
(a)(ii)	$\sum_{r=1}^n p_r = \sum_{r=1}^n \frac{1}{3}(7 - 4^r) \quad (\text{from (i)})$ $= \frac{1}{3} \left[\sum_{r=1}^n 7 - \sum_{r=1}^n 4^r \right]$ $= \frac{1}{3} \left[7n - \frac{4(1 - 4^n)}{1 - 4} \right]$ $= \frac{1}{3} \left[7n + \frac{4}{3}(1 - 4^n) \right]$
(b)(i)	As $n \rightarrow \infty$, $\frac{1}{(n+1)!} \rightarrow 0$. Therefore, $\sum u_r$ converges and the sum to infinity is 1.
(b)(ii)	$u_n = S_n - S_{n-1}$ $= 1 - \frac{1}{(n+1)!} - \left[1 - \frac{1}{((n-1)+1)!} \right]$ $= -\frac{1}{(n+1)!} + \frac{1}{n!}$ $= \frac{-1 + n + 1}{(n+1)!} \quad \frac{1}{n!} = \frac{n+1}{(n+1)!}$ $= \frac{n}{(n+1)!}$

7	Suggested Solutions
(i)	Using GC, $\alpha = 1.885$ (3 d.p.) To find β, $-7 = x^6 - 3x^4 - 7$ $x^6 - 3x^4 = 0$ $x^4(x^2 - 3) = 0$ $x = 0 \text{ or } x = -\sqrt{3} \text{ or } x = \sqrt{3}$ Since $\beta > 0$, $\beta = \sqrt{3}$.
(ii)	Using GC, $\int_{\beta}^{\alpha} f(x) dx = \int_{\sqrt{3}}^{1.88525} x^6 - 3x^4 - 7 dx$ $= -0.597 \text{ (3 d.p.)}$
(iii)	Required Area $= \int_0^{\sqrt{3}} -7 - (x^6 - 3x^4 - 7) dx$ Let $y_1 = -7$ and $y_2 = x^6 - 3x^4 - 7$. Required area $= \int_0^{\sqrt{3}} y_1 - y_2 dx$ $= - \left[\frac{x^7}{7} - \frac{3x^5}{5} \right]_0^{\sqrt{3}}$ $= - \left(\frac{(\sqrt{3})^7}{7} - \frac{3(\sqrt{3})^5}{5} \right)$ $= - \frac{27\sqrt{3}}{7} + \frac{27\sqrt{3}}{5}$ $= \frac{54}{35} \sqrt{3}$
(iv)	$f(x) = x^6 - 3x^4 - 7$ $f(-x) = (-x)^6 - 3(-x)^4 - 7$ $= x^6 - 3x^4 - 7$ $= f(x)$ $\Rightarrow f(x) = f(-x)$ Since $f(x) = f(-x)$, the only two real roots of $f(x) = 0$ are 1.885 and -1.885. (From (i)) Since $f(x)$ is a polynomial with real coefficients, the remaining 4 complex roots are in a form of 2 complex conjugate pairs.

8	Suggested Solutions
(i)	$\int \frac{1}{\sqrt{9-x^2}} dx = \sin^{-1}\left(\frac{x}{3}\right) + C, C \in \mathbb{R}$
(ii)	$\begin{aligned} & \frac{1}{\sqrt{9-x^2}} \\ &= (9-x^2)^{-\frac{1}{2}} \\ &= 9^{-\frac{1}{2}} \left(1 - \frac{x^2}{9}\right)^{-\frac{1}{2}} \\ &= \frac{1}{3} \left(1 - \frac{x^2}{9}\right)^{-\frac{1}{2}} \\ &= \frac{1}{3} \left(1 + \left(-\frac{1}{2}\right) \left(-\frac{x^2}{9}\right) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!} \left(-\frac{x^2}{9}\right)^2 + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!} \left(-\frac{x^2}{9}\right)^3 + \dots \right) \\ &= \frac{1}{3} + \frac{1}{54}x^2 + \frac{1}{648}x^4 + \frac{5}{34992}x^6 + \dots \quad (\text{up to } x^6) \end{aligned}$
(iii)	Note that $\int \frac{1}{\sqrt{9-x^2}} dx = \sin^{-1}\left(\frac{x}{3}\right) + C$ from (i). $\frac{1}{\sqrt{9-x^2}} = \frac{1}{3} + \frac{1}{54}x^2 + \frac{1}{648}x^4 + \frac{5}{34992}x^6 + \dots \quad (\text{from (ii)})$ $\int \frac{1}{\sqrt{9-x^2}} dx = \int \left(\frac{1}{3} + \frac{1}{54}x^2 + \frac{1}{648}x^4 + \frac{5}{34992}x^6 + \dots \right) dx$ $\sin^{-1}\left(\frac{x}{3}\right) + D = \frac{1}{3}x + \frac{1}{162}x^3 + \frac{1}{3240}x^5 + \frac{5}{244944}x^7 + \dots, D \in \mathbb{R}$ When $x = 0$, $D = 0$. $\therefore \sin^{-1}\left(\frac{x}{3}\right) = \frac{1}{3}x + \frac{1}{162}x^3 + \frac{1}{3240}x^5 + \frac{5}{244944}x^7 + \dots$

9	Suggested Solutions
(i)	Equation of plane p: $x + 2y - 3z = 12 \Rightarrow \mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 12$ Equation of line l: $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ $\mathbf{r} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}, \lambda \in \mathbb{R}$ Let $\lambda = \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}, \lambda \in \mathbb{R}.$ $x = 1 + 2\lambda$ $y = -1 - \lambda$ $z = 3 + 4\lambda$ Since planes p and q are perpendicular, $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix}$ is parallel to plane q. $\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ -10 \\ -5 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ $\Rightarrow \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ is a normal vector of plane q. Equation of plane q: $\mathbf{r} \cdot \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ $= 1 + 2 - 3 = 0$ $x - 2y - z = 0$ Must remember to answer question and leave in Cartesian form.
(ii)	$x + 2y - 3z = 12 \quad \text{--- (1)}$ $x - 2y - z = 0 \quad \text{--- (2)}$ Solving (1) and (2) using GC, $x = 6 + 2\mu$ $y = 3 + \frac{1}{2}\mu, \mu \in \mathbb{R}$ $z = \mu$ Note: use a different parameter from the one used in (i)

	Equation of line m : $\mathbf{r} = \begin{pmatrix} 6 \\ 3 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1/2 \\ 1 \end{pmatrix}, \mu \in \mathbb{R}$
(iii)	Since B is a general point on m, $\overrightarrow{OB} = \begin{pmatrix} 6+2\mu \\ 3+(1/2)\mu \\ \mu \end{pmatrix} \text{ for some } \mu \in \mathbb{R}.$ $\overrightarrow{AB} = \begin{pmatrix} 6+2\mu \\ 3+(1/2)\mu \\ \mu \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5+2\mu \\ 4+(1/2)\mu \\ \mu-3 \end{pmatrix}$ $\therefore \overrightarrow{AB} ^2 = (5+2\mu)^2 + \left(4 + \frac{1}{2}\mu\right)^2 + (\mu-3)^2$ $\frac{d(\overrightarrow{AB} ^2)}{d\mu} = 2(5+2\mu)(2) + 2\left(4 + \frac{1}{2}\mu\right)\left(\frac{1}{2}\right) + 2(\mu-3)$ $= \frac{21}{2}\mu + 18$ $\frac{d(\overrightarrow{AB} ^2)}{d\mu} = 0$ $10.5\mu + 18 = 0$ $\mu = -\frac{12}{7}$ $\frac{d^2(\overrightarrow{AB} ^2)}{d\mu^2} = 10.5 > 0$ $\therefore$ coordinates of point on m which is nearest to A $= \left(6 + 2\left(-\frac{12}{7}\right), 3 + \frac{1}{2}\left(-\frac{12}{7}\right), -\frac{12}{7}\right)$ $= \left(\frac{18}{7}, \frac{15}{7}, -\frac{12}{7}\right)$ <u>Alternative Method</u> For B nearest to A, B will be the foot of perpendicular from A to the line m. B nearest to A $\Rightarrow$ finding minimum $\overrightarrow{AB}$ $\Rightarrow$ finding minimum $\overrightarrow{AB} ^2$ $\Rightarrow \frac{d \overrightarrow{AB} ^2}{d\mu} = 0$

	$\overrightarrow{AB} \cdot \begin{pmatrix} 2 \\ 1/2 \\ 1 \end{pmatrix} = 0$ $\begin{pmatrix} 5+2\mu \\ 4+(1/2)\mu \\ \mu-3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1/2 \\ 1 \end{pmatrix} = 0$ $10 + 4\mu + 2 + \frac{1}{4}\mu - 3 + \mu = 0$ $\mu = -\frac{12}{7}$ $\therefore$ coordinates of point on m which is nearest to A $= \left(6 + 2\left(-\frac{12}{7}\right), 3 + \frac{1}{2}\left(-\frac{12}{7}\right), -\frac{12}{7} \right)$ $= \left(\frac{18}{7}, \frac{15}{7}, -\frac{12}{7} \right)$
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10	Suggested Solutions
(i)	When $x = \frac{1}{2}$ and $\frac{dx}{dt} = -\frac{1}{4}$, $-\frac{1}{4} = k \left(1 + \frac{1}{2} - \left(\frac{1}{2} \right)^2 \right)$ $-\frac{1}{4} = \frac{5}{4}k$ $\therefore k = -\frac{1}{5} \text{ (shown)}$
(ii)	$1 + x - x^2 = -(x^2 - x - 1)$ $= -\left[\left(x - \frac{1}{2} \right)^2 - \frac{5}{4} \right]$ $= \frac{5}{4} - \left(x - \frac{1}{2} \right)^2$

	$\frac{dx}{dt} = -\frac{1}{5}(1+x-x^2)$ $\int \frac{1}{1+x-x^2} dx = \int -\frac{1}{5} dt$ $\int \frac{1}{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(x - \frac{1}{2}\right)^2} dx = \int -\frac{1}{5} dt$ $\frac{1}{\sqrt{5}} \ln \left \frac{\frac{\sqrt{5}}{2} + \left(x - \frac{1}{2}\right)}{\frac{\sqrt{5}}{2} - \left(x - \frac{1}{2}\right)} \right = -\frac{1}{5}t + c, c \in \mathbb{R}$ $\frac{1}{\sqrt{5}} \ln \left(\frac{\frac{\sqrt{5}}{2} + \left(x - \frac{1}{2}\right)}{\frac{\sqrt{5}}{2} - \left(x - \frac{1}{2}\right)} \right) = -\frac{1}{5}t + c \quad \left(\because 0 \leq x \leq \frac{1}{2} \right)$ $\frac{1}{\sqrt{5}} \ln \left(\frac{\sqrt{5} + 2x - 1}{\sqrt{5} - 2x + 1} \right) = -\frac{1}{5}t + c$ When $t = 0$ and $x = \frac{1}{2}$, $c = 0$. $\frac{1}{\sqrt{5}} \ln \left(\frac{\sqrt{5} + 2x - 1}{\sqrt{5} - 2x + 1} \right) = -\frac{1}{5}t$ $\therefore t = -\sqrt{5} \ln \left(\frac{\sqrt{5} + 2x - 1}{\sqrt{5} - 2x + 1} \right).$
(iii)(a)	When $x = \frac{1}{4}$, $t = -\sqrt{5} \ln \left(\frac{\sqrt{5} + \frac{1}{2} - 1}{\sqrt{5} - \frac{1}{2} + 1} \right)$ $t = -\sqrt{5} \ln \left(\frac{\sqrt{5} - \frac{1}{2}}{\sqrt{5} + \frac{1}{2}} \right) = -\sqrt{5} \ln \left(\frac{2\sqrt{5} - 1}{2\sqrt{5} + 1} \right)$
(iii)(b)	When $x = 0$, $t = -\sqrt{5} \ln \left(\frac{\sqrt{5} - 1}{\sqrt{5} + 1} \right) = 2.152 \text{ (3 d.p.)}$

(iv)

$$t = -\sqrt{5} \ln \left(\frac{\sqrt{5} + 2x - 1}{\sqrt{5} - 2x + 1} \right)$$

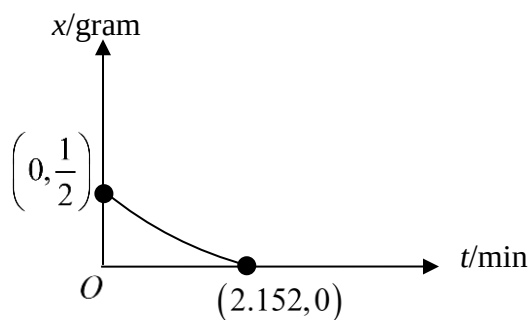
$$\frac{\sqrt{5} + 2x - 1}{\sqrt{5} - 2x + 1} = e^{-\frac{t}{\sqrt{5}}}$$

$$\sqrt{5} + 2x - 1 = \sqrt{5}e^{-\frac{t}{\sqrt{5}}} - 2xe^{-\frac{t}{\sqrt{5}}} + e^{-\frac{t}{\sqrt{5}}}$$

$$2x + 2xe^{-\frac{t}{\sqrt{5}}} = \sqrt{5}e^{-\frac{t}{\sqrt{5}}} + e^{-\frac{t}{\sqrt{5}}} - \sqrt{5} + 1$$

$$2x \left(1 + e^{-\frac{t}{\sqrt{5}}} \right) = \sqrt{5}e^{-\frac{t}{\sqrt{5}}} + e^{-\frac{t}{\sqrt{5}}} - \sqrt{5} + 1$$

$$x = \frac{\sqrt{5}e^{-\frac{t}{\sqrt{5}}} + e^{-\frac{t}{\sqrt{5}}} - \sqrt{5} + 1}{2 \left(1 + e^{-\frac{t}{\sqrt{5}}} \right)}$$



11	Suggested Solutions
(i)	$V = \frac{1}{3}\pi r^2 h + \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3 \text{ --- (1)}$ By Pythagoras Theorem, $h^2 + r^2 = 4^2 \Rightarrow h = \sqrt{16 - r^2}$ --- (2) (since $h > 0$) Solving (1) and (2), $V = \frac{1}{3}\pi r^2 \sqrt{16 - r^2} + \frac{2}{3}\pi r^3$ $\frac{dV}{dr} = \frac{1}{3}\pi \left(2r\sqrt{16 - r^2} + r^2 \left(\frac{1}{2}(16 - r^2)^{-\frac{1}{2}}(-2r) \right) \right) + \frac{2}{3}(3\pi r^2)$ $= \frac{1}{3}\pi \left(2r\sqrt{16 - r^2} - \frac{r^3}{\sqrt{16 - r^2}} \right) + 2\pi r^2$ For maximum V, $\frac{dV}{dr} = 0$. $\frac{1}{3}\pi \left(2r\sqrt{16 - r^2} - \frac{r^3}{\sqrt{16 - r^2}} \right) + 2\pi r^2 = 0$ $2\sqrt{16 - r^2} - \frac{r^2}{\sqrt{16 - r^2}} + 6r = 0 \quad (\text{since } r \neq 0)$ $2(16 - r^2) - r^2 + 6r\sqrt{16 - r^2} = 0$ $6r\sqrt{16 - r^2} = 3r^2 - 32$ $(6r\sqrt{16 - r^2})^2 = (3r^2 - 32)^2$ $36r^2(16 - r^2) = 9r^4 - 192r^2 + 1024$ $576r^2 - 36r^4 = 9r^4 - 192r^2 + 1024$ $45r^4 - 768r^2 + 1024 = 0 \quad (\text{shown})$ Therefore, $r = r_1$ must satisfy $45r^4 - 768r^2 + 1024 = 0$.
(ii)	Using GC, for $r > 0$, $r = 3.95080$ or $r = 1.20742$ $= 3.951 \text{ (3 d.p.)}$ $= 1.207 \text{ (3 d.p.)}$
(iii)	For $r = 3.95080$, $\frac{dV}{dr} = \frac{1}{3}\pi \left(2(3.95080)\sqrt{16 - 3.95080^2} - \frac{3.95080^3}{\sqrt{16 - 3.95080^2}} \right) + 2\pi(3.95080)^2$ $= -0.00867 \approx 0$ For $r = 1.20742$, $\frac{dV}{dr} = \frac{1}{3}\pi \left(2(1.20742)\sqrt{16 - 1.20742^2} - \frac{1.20742^3}{\sqrt{16 - 1.20742^2}} \right) + 2\pi(1.20742)^2$ $= 18.320 \neq 0$

	Only $r = 3.951$ gives a stationary value of $V \Rightarrow r_1 = 3.951$. For $r_1 = 3.95080$, $h = \sqrt{16 - 3.95080^2}$ $= 0.625 \text{ (3 s.f)}$
(iv)	The graph shows a curve starting at the origin (0,0) and increasing to a point at (4,134). A stationary point is marked at (3.951,139). The axes are labeled V and r, with the origin O. A blue box contains instructions: 'Use Window Setting with Xmin = 0 and Xmax = 4 (from context of question). Then use 0:Zoomfit to get the full graph.'